



The helix slows down the propagation of electrons as these travel down the tube. The electrons bunch, and reinforce the voltage in the helix, which creates amplification (Courtesy: Thales Electron Devices)

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Microwave tubes have potential applications in radar, electronic warfare and communication systems. Air-traffic-control radars, military radars, ground penetrating radars, imaging radars, UWB radars, cloud radars and

space debris radars are some types of radars that use microwave tubes. Multi-beam jammers, phase-array jammers, and ultra-high-frequency and ultra-wide-bandwidth jammers used in electronic warfare also use microwave tubes. Clouds can be seen by microwave. These tubes play a role in climate forecasting and some medical applications (used in the diagnosis of hyperthermia) as well. Besides, microwave has its utilities for common man in the form of microwave heating and microwave protective gear/wall paper/furnishing.

Salient features of microwave-based devices

Parameter	Value
High power	mW-GW
High efficiency	10-80 per cent
High gain	60 dB
Frequency range	1-1000 GHz
Broad bandwidth	3 octaves

Basic principle of microwave devices

The efficiency of conventional tubes is largely independent of the frequency up to a certain limit. When the frequency increases beyond that limit, several factors combine to rapidly decrease the tube's efficiency. The high-frequency effects in conventional tubes are circuit

Microwave devices

Conventional devices. Klystrons, magnetrons, traveling-wave tubes, backward-wave oscillators and crossed-field amplifiers

New-generation devices. Cyclotron resonance devices (gyrotrons, gyro-TWTs and klystron-BWOs), Cerenkov radiation devices (magnetrons/BWOs/TWTs, orotrons, magnetically insulated line oscillators), Doppler effect-based devices (free-electron lasers, ubitrons, cyclotron auto-resonance masers), space-charge devices (virtual cathode oscillators) and multi-beam devices

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